

Experiment No: 7

AIM: Implement the embedding wallet (Metamask) and transaction using Solidity

Theory:

Blockchain is a decentralized and distributed digital ledger that records transactions securely across multiple systems. It ensures transparency, security, and immutability, meaning once data is recorded, it cannot be changed.

Ethereum is a popular blockchain platform that allows the execution of Smart Contracts. Smart contracts are self-executing programs written in Solidity that automatically perform actions when predefined conditions are met, eliminating the need for intermediaries.

MetaMask is a cryptocurrency wallet and browser extension used to interact with blockchain networks. It stores user accounts and private keys securely and allows users to send, receive, and sign transactions. It also acts as a bridge between Remix IDE and the blockchain network.

Testnets such as Sepolia and RSK Testnet are used for testing purposes. They provide free tokens, allowing developers to deploy and test smart contracts without using real cryptocurrency, reducing financial risk.

Remix IDE is a web-based development environment used to write, compile, deploy, and test smart contracts. It provides a user-friendly interface and supports direct integration with MetaMask for deploying contracts on blockchain networks.

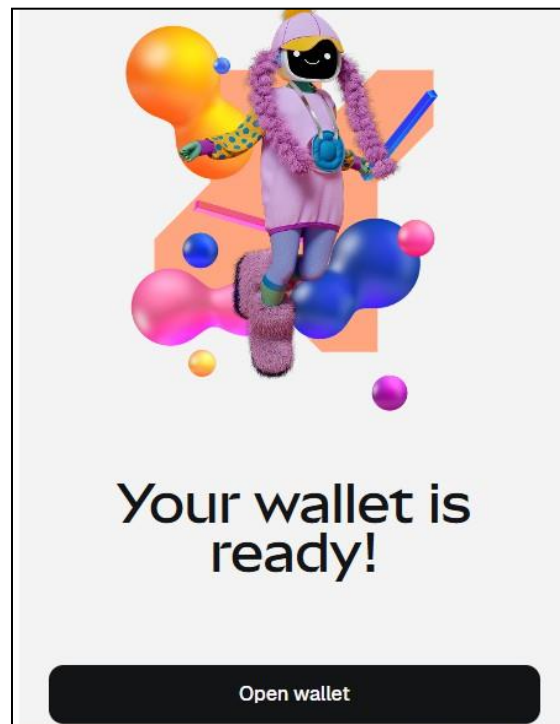
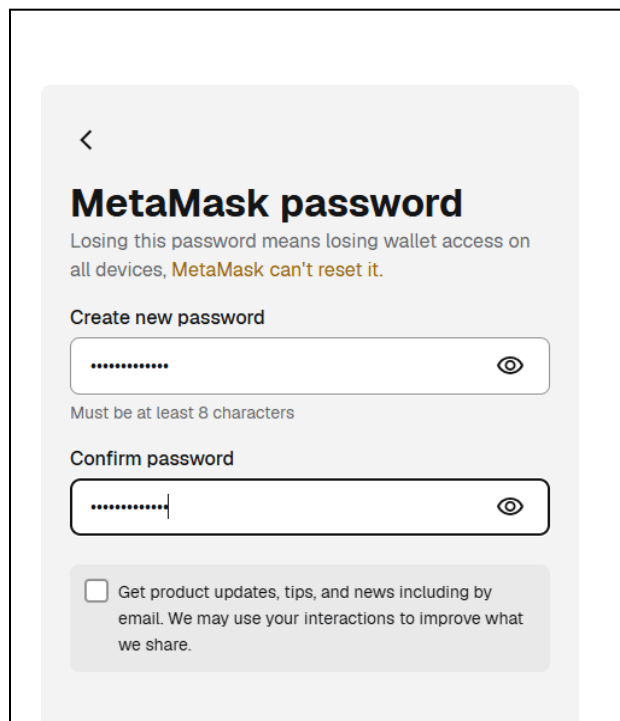
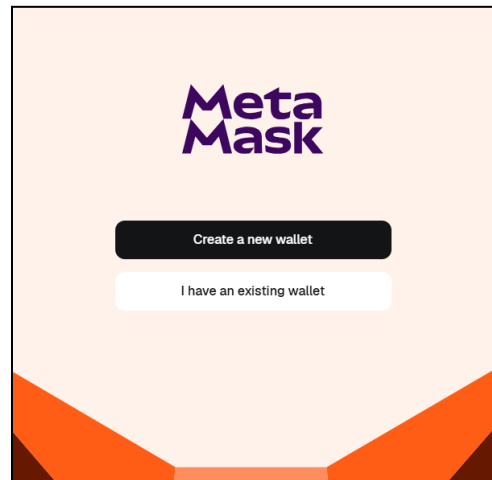
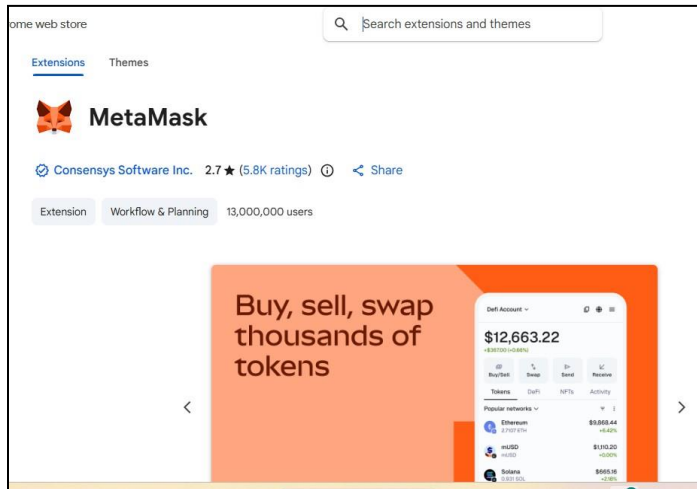
The smart contract deployment process includes writing the contract in Solidity, compiling it to remove errors, connecting MetaMask to Remix, and deploying the contract on a testnet by paying gas fees using test tokens.

A blockchain transaction refers to any interaction with the network, such as deploying a contract or calling its functions. Each transaction requires authentication through a digital signature and is permanently recorded on the blockchain. These transactions can be verified using blockchain explorers.

Implementation

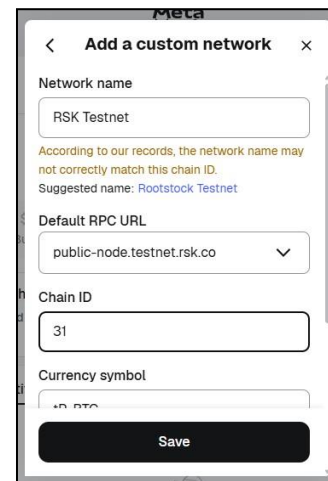
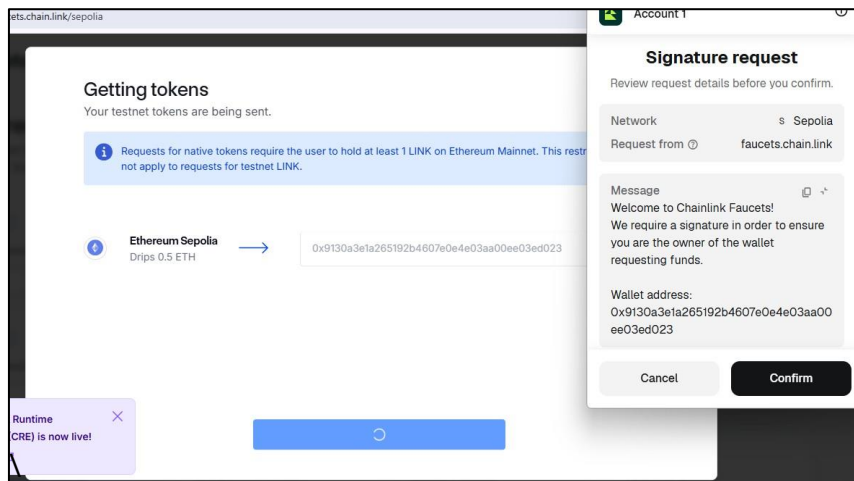
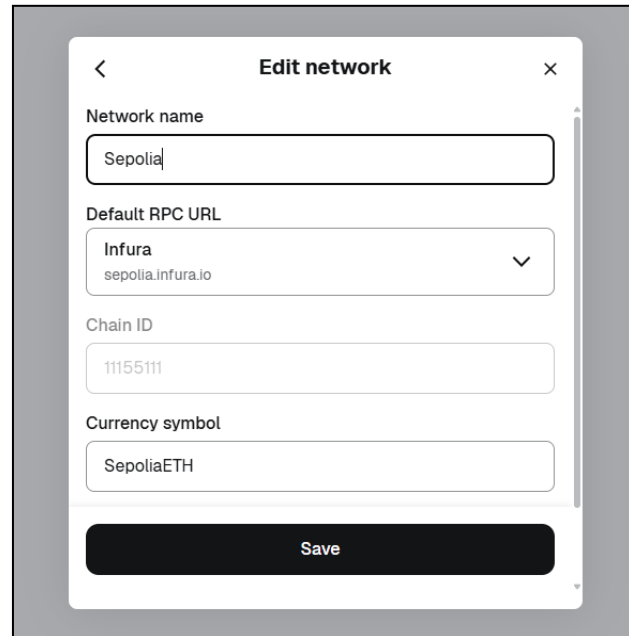
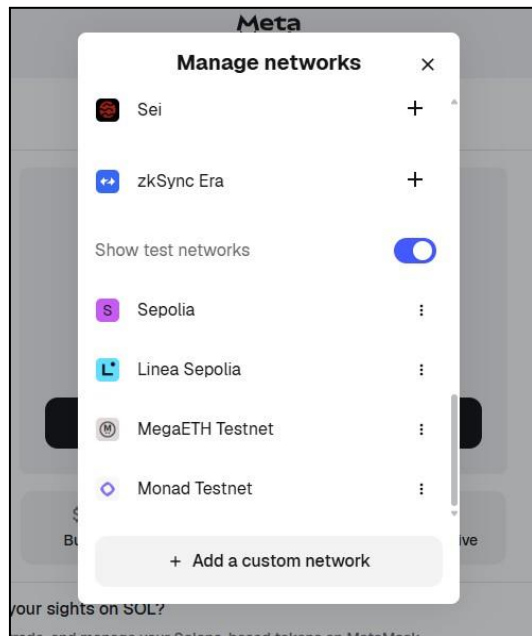
Step 1: Install MetaMask

Install MetaMask extension and create a wallet



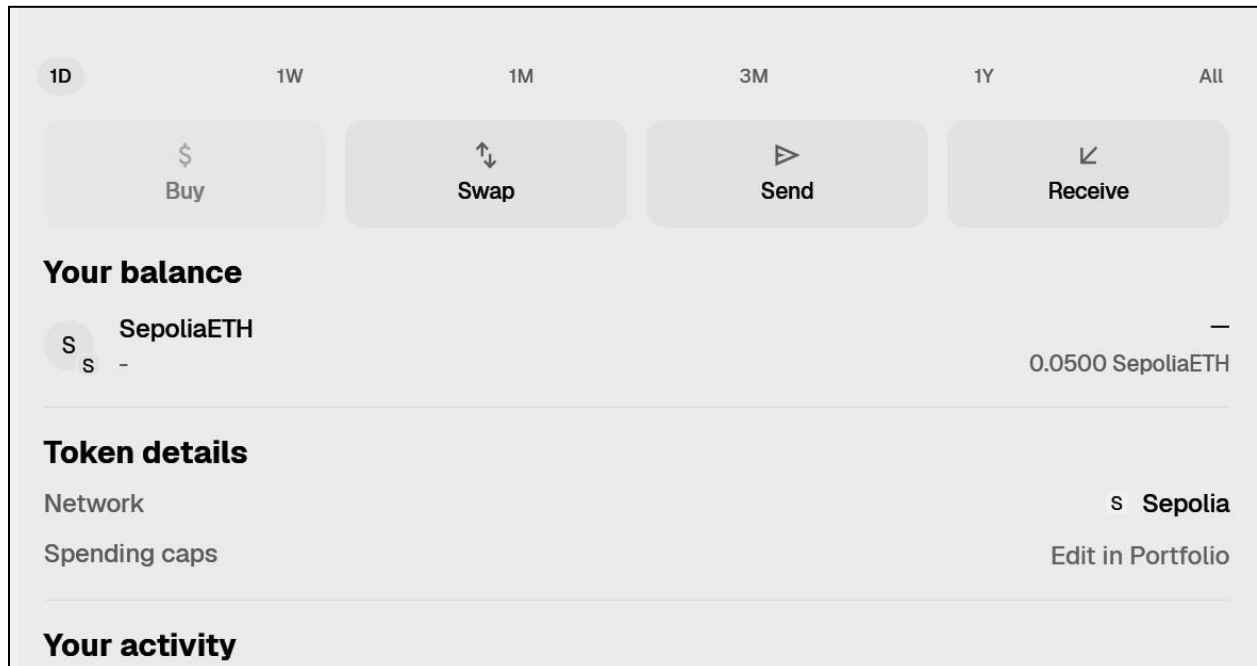
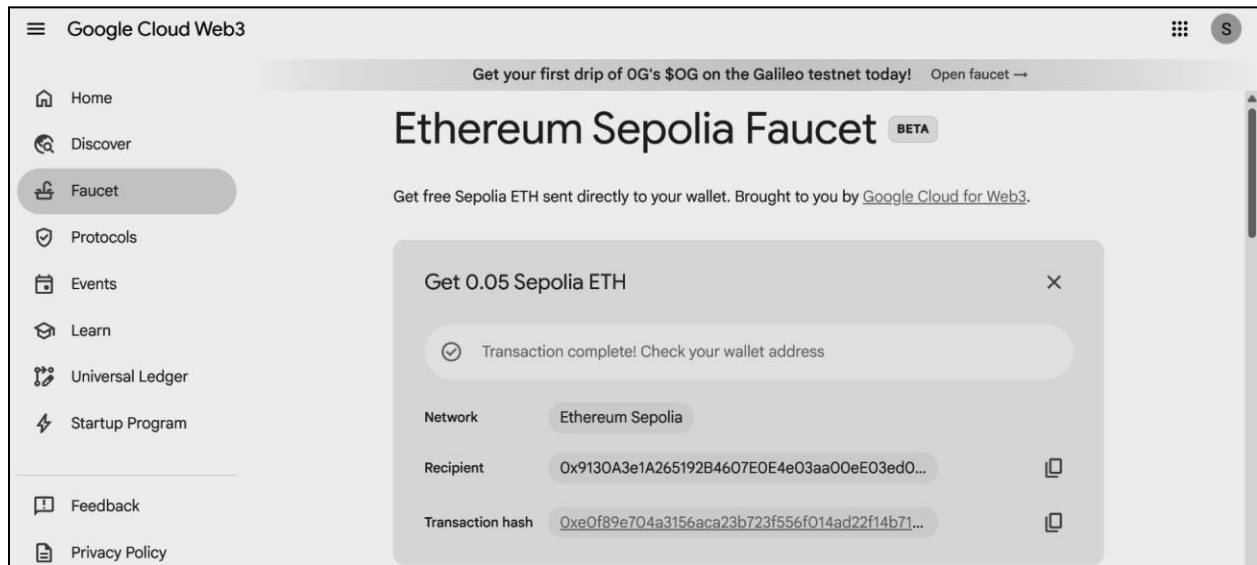
Step 2: Add Sepolia Testnet

Configure MetaMask to connect to Sepolia network.



Step 3: Fund Wallet

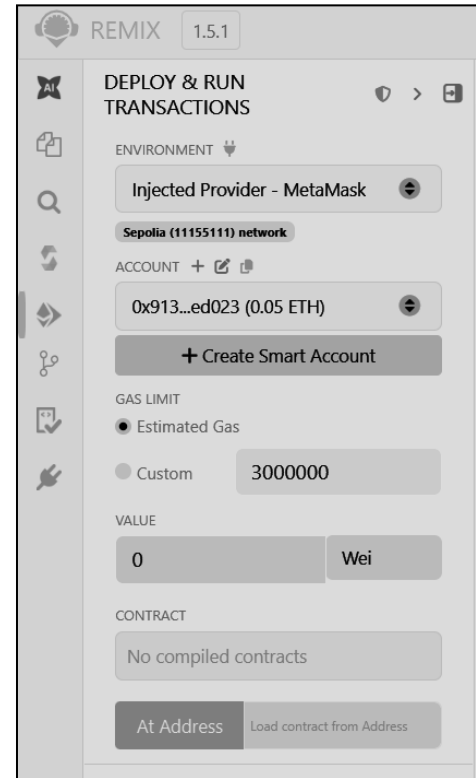
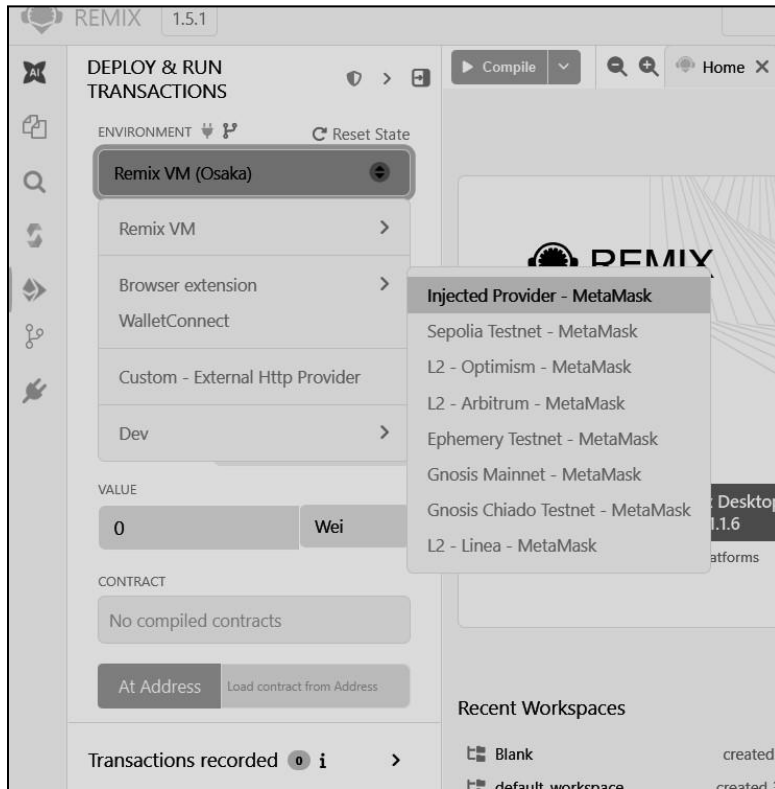
Use faucet to get test ETH



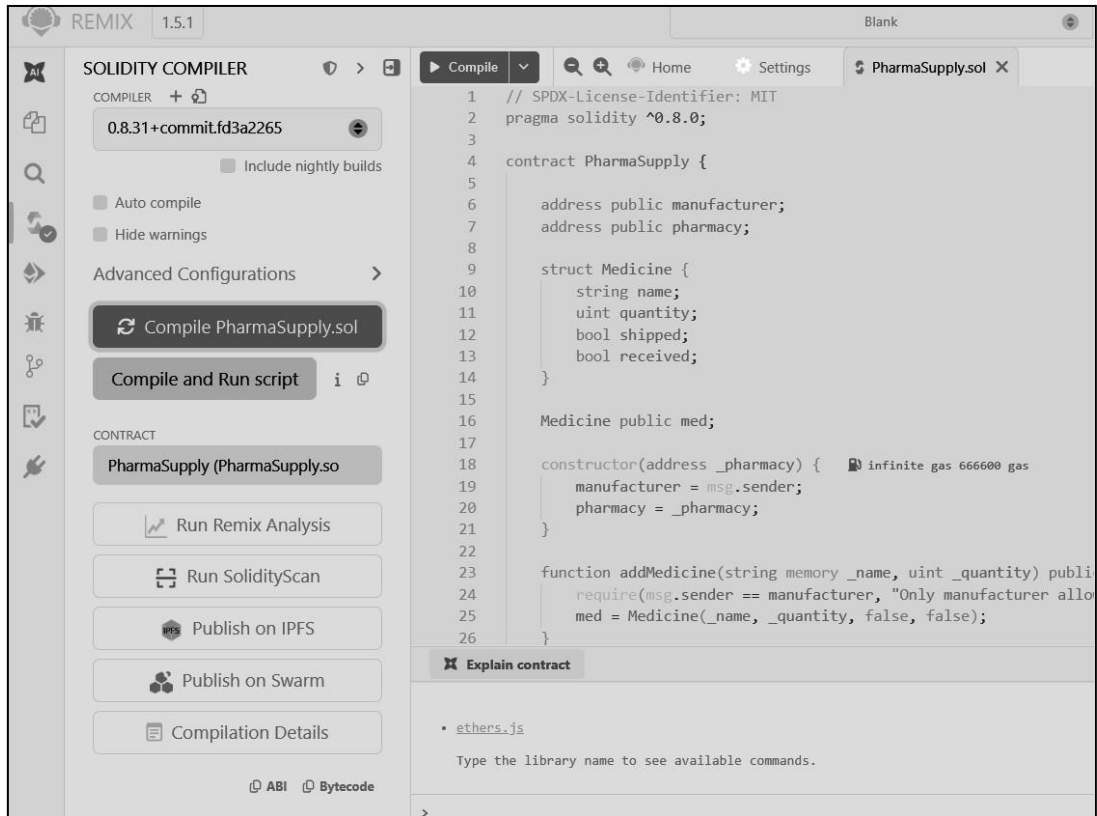
Step4: Connect the Sepolia Testnet / RSK Testnet to Remix IDE

1. In the Remix IDE, select the Environment as Injected Web3.

Note : Injected Web3 connects Remix with the active account in Metamask.

**Compile the Contract**

1. Open Remix
2. Paste your contract code
3. Go to Solidity Compiler (left panel)
4. Select version 0.8.x (match ^0.8.0)
5. Click Compile PharmaSupply.sol

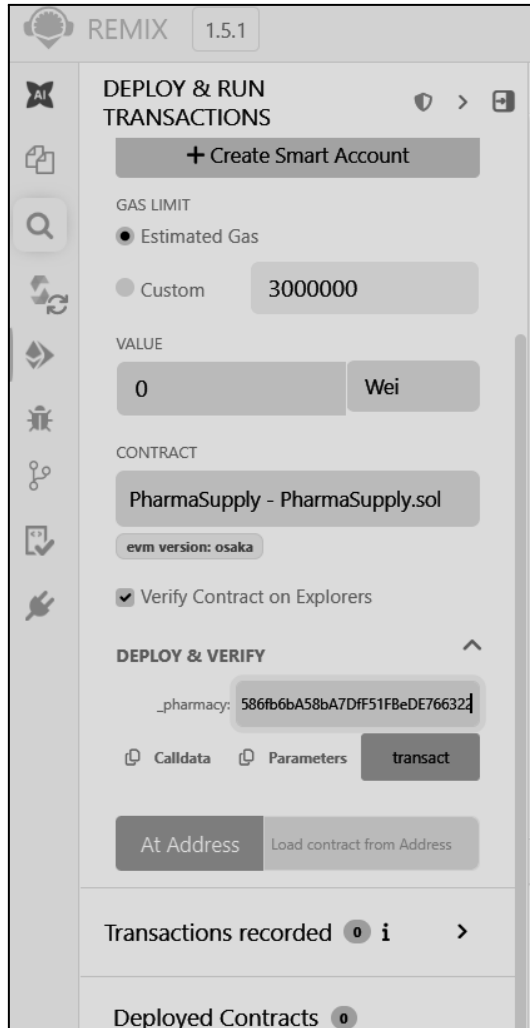


Connect MetaMask (Injected Provider)

1. Go to Deploy & Run Transactions tab
2. In Environment → select Injected Provider - MetaMask
3. MetaMask popup will open → click Connect

Make sure:

- Network = Sepolia Test Network
- You have 0.05 Sepolia ETH



Deploy Contract

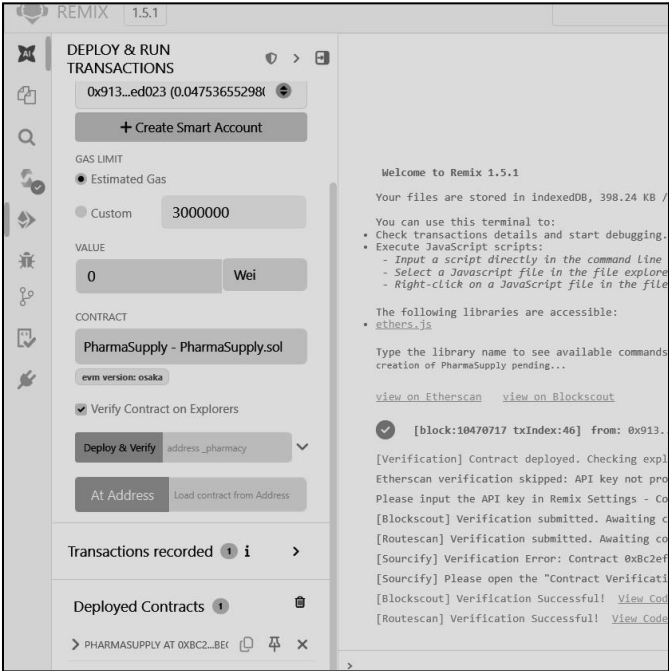
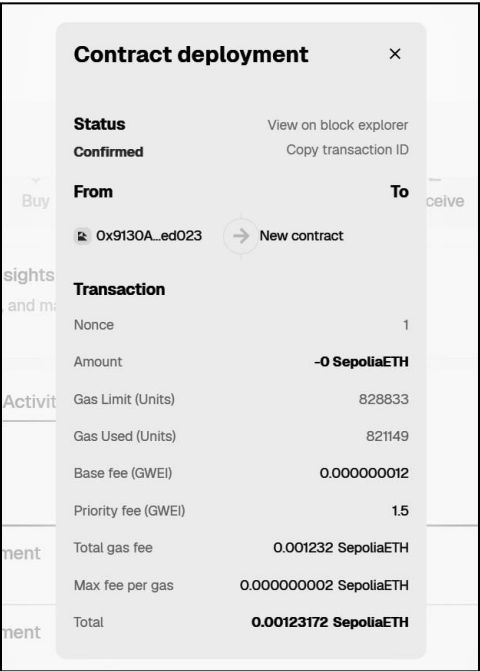
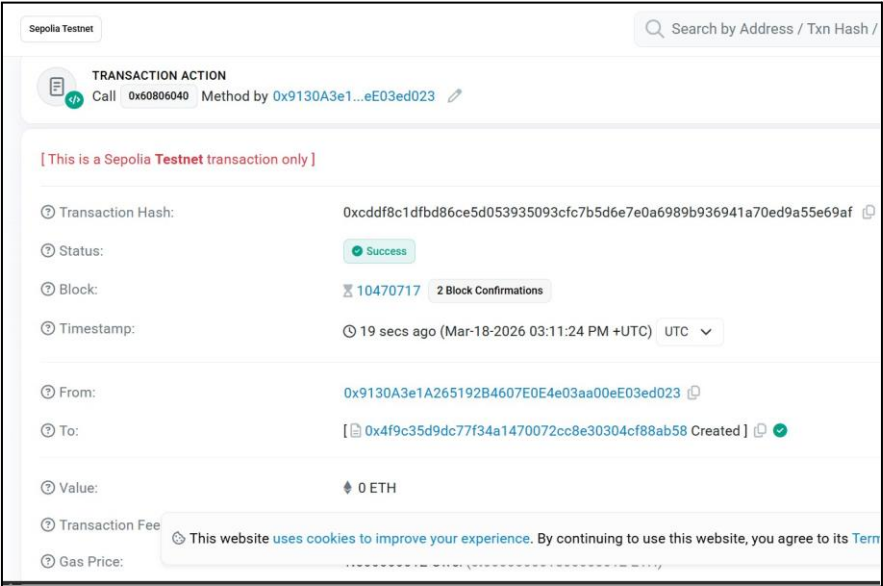
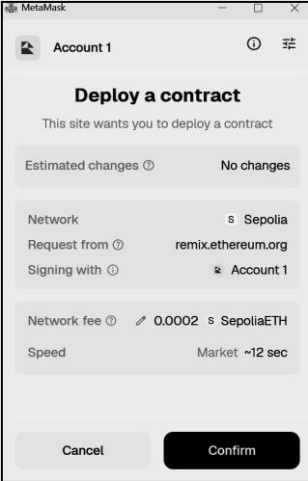
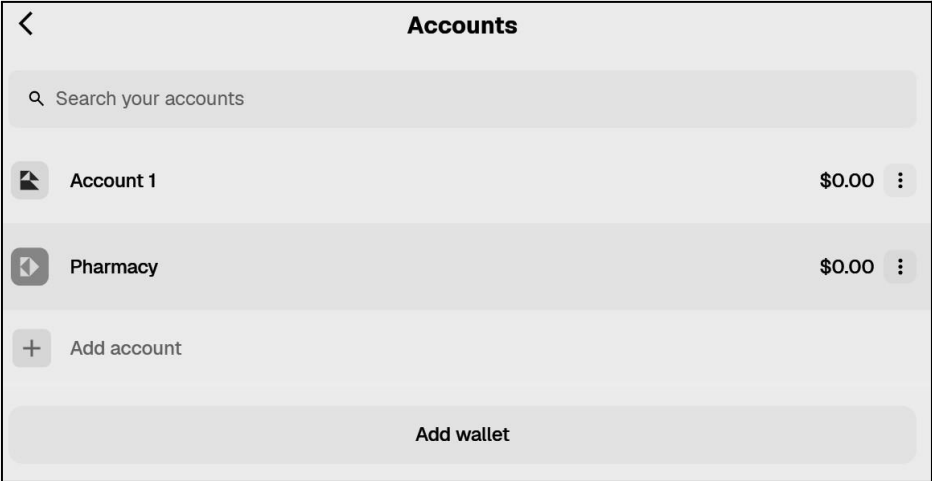
Your constructor needs a pharmacy address

Steps:

1. Copy your second MetaMask account address (pharmacy)
 - MetaMask → Account 2 → Copy address
2. Paste that address in constructor field
3. Click Deploy
4. MetaMask will pop up → Confirm transaction

Wait a few seconds...

After success → your contract appears under Deployed Contracts

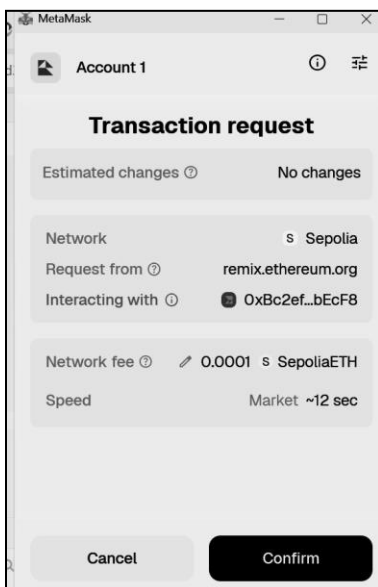
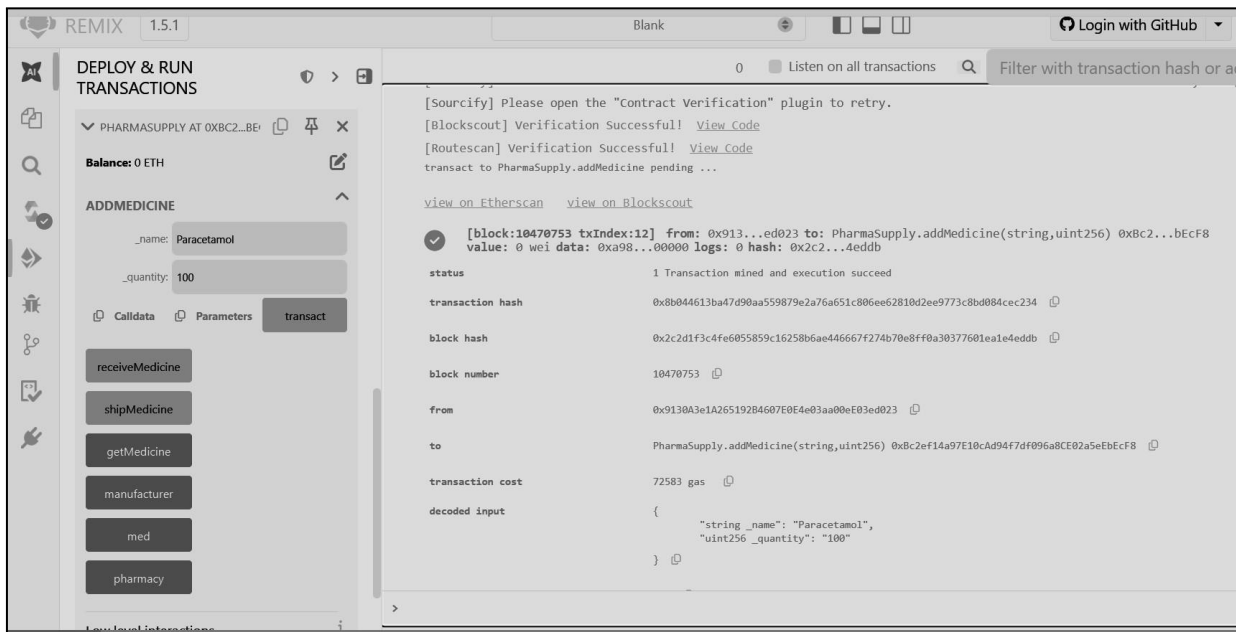


Interact with Contract

Step 1: Add Medicine (Manufacturer only)

- Make sure MetaMask is on Account 1 (manufacturer)
- In Remix:
 - Enter:
 - `_name`: "Paracetamol"
 - `_quantity`: 100
- Click addMedicine

Confirm in MetaMask



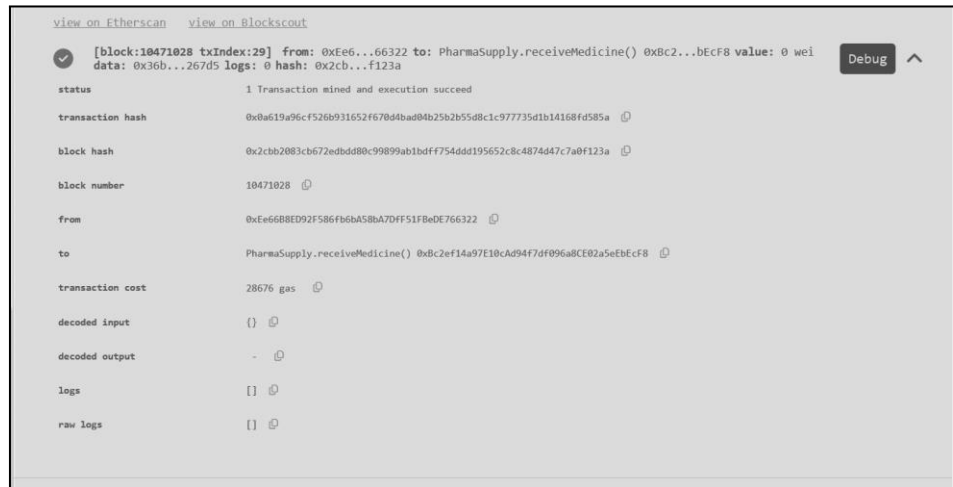
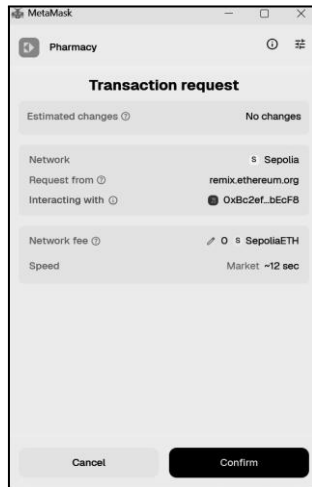
- Still on Account 1
- Click shipMedicine
- Confirm transaction

The image is a screenshot of the MetaMask mobile application interface. At the top, the status bar shows the time as 12:42 and the battery level at 92%. The app's header bar displays the MetaMask logo and the account name 'Account 1'. To the right of the account name are two icons: a circular arrow and a grid of squares. The main heading of the screen is 'Transaction request'. Below this, there are two sections. The first section, 'Estimated changes', shows 'No changes' on the right. The second section, 'Network', shows 's Sepolia'. Below the network section, there are three rows of information: 'Request from' with the address 'remix.ethereum.org', 'Interacting with' with a circular icon and the address '0xBc2ef...bEcF8', and 'Network fee' with a pencil icon, the value '0.0001', 's SepoliaETH', and 'Speed' with 'Market ~12 sec'. At the bottom of the screen, there are two large buttons: 'Cancel' on the left and 'Confirm' on the right.

[illegible]

Step 3: Receive Medicine (Pharmacy)

- Switch MetaMask to Account 2 (pharmacy)
- Click receiveMedicine
- Confirm transaction



Step 4: Check Data

- Click getMedicine

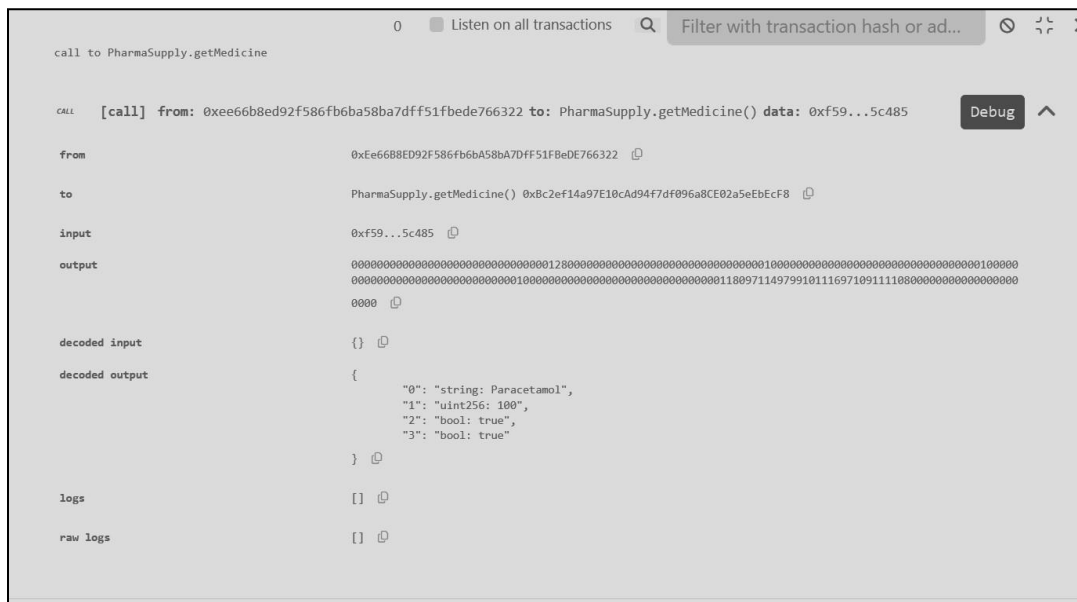
Output will be like:

"Paracetamol"

100

true (shipped)

true (received)



Compiled PharmaSupply.sol

Copy contract address → paste in Etherscan (Sepolia) See all transactions + status

The screenshot shows the Etherscan Sepolia Testnet interface. At the top, there's a search bar and navigation links. The main section displays the contract address `0xBc2ef14a97E10cAd94f7df096a8CE02a5eEbEcF8`. Below this, there are three tabs: Overview, More Info, and Multichain Info. The Overview tab shows the ETH balance as 0 ETH. The More Info tab shows the contract creator as `0x9130A3e1...eE03ed023` with a timestamp of 1 hr ago. The Multichain Info tab shows N/A.

The screenshot shows the Transactions tab of the Etherscan interface. It displays a list of transactions for the contract address `0xBc2ef14a97E10cAd94f7df096a8CE02a5eEbEcF8`. The table includes columns for Transaction Hash, Method, Block, Age, From, To, Amount, and Txn Fee. There are three transactions listed, all with an amount of 0 ETH.

Transaction Hash	Method	Block	Age	From	To	Amount	Txn Fee
0x0a619a96cf5...	<code>0x36b267d5</code>	10471028	5 mins ago	0xEe66B8ED...eDE766322	0xBc2ef14a...a5eEbEcF8	0 ETH	0.00004301
0x2c42b81faf5...	<code>0x39f32075</code>	10470765	1 hr ago	0x9130A3e1...eE03ed023	0xBc2ef14a...a5eEbEcF8	0 ETH	0.00006835
0x8b044613ba...	<code>Add Medicine</code>	10470753	1 hr ago	0x9130A3e1...eE03ed023	0xBc2ef14a...a5eEbEcF8	0 ETH	0.00010887

Conclusion

The experiment successfully demonstrated MetaMask integration with Remix IDE and execution of blockchain transactions. It provided hands-on experience with smart contracts, ensuring secure and transparent interactions on a testnet.